Let the AI break the lab.

Network as Code in the Real World



Patrick Mosimann
CISCO · SOLUTIONS ENGINEER

Dawid Bednarczyk
CISCO CX · DELIVERY ARCHITECT

CHNUG #5 · 12 NOV 2026 · CISCO WALLISELLEN

PM

AI IS COMING TO YOUR NETWORK

Would you give an AI enable on your core switch?

Neither would we.

No AI ever logs into the customer's production network.

But it writes almost every change.

Data model, Terraform fixes, tests, capture scripts, plans, and the meeting notes.

The difference: code.

Every AI idea becomes a **diff**. In steps: validated, tested, approved by people.

cisco

PM

TODAY · 30 MINUTES

One idea, four chapters

01 · PATRICK

The method

Network as Code, the toolchain, and how it works for any vendor.

02 · PATRICK

Demo

From a prompt to a running network, through Git.

03 · DAWID

A real rollout

A European insurer, all as code, with AI in every step.

04 · BOTH

Monday morning

Six lessons, where to start, your questions.

cisco

PM

01 · WHY CODE

Most outages are not hardware. They are us.

80%+

of network problems come from misconfiguration and change management

Gartner, 2024

25%

of TAC cases relate to misconfiguration

Forbes, 2025

33%

of organisations automate more than a quarter of their network work

Gartner, 2025

`no version control → drift`

`no automation → snowflakes`

`no testing → outages`

cisco

PM

01 · NETWORK AS CODE

Describe the network in YAML. Let the pipeline do the rest.



Declare
YAML data model in Git

Validate
syntax, schema, business rules
`nac-validate`

Plan + review
terraform plan, pull request / ITSM
`terraform`

Deploy
tested, maintained Terraform modules
`nac-* modules`

Test
tests generated from the same data model
`nac-test`

"No need to learn a new language: use text to describe your network." Same method across **ACI, VXLAN, NX-OS, IOS-XE, SD-WAN, Catalyst Center, ISE, Meraki, FMC**, and more.

`open-source toolchain`

`built by Cisco CX`

source: netascode.cisco.com

cisco

PM

01 · ONE LEVEL LOWER

One repo, one merge request, five stages

THE REPO

```
nac-catalystcenter/
├─ data/              # the network, in YAML
│  ├─ sites.nac.yaml
│  └─ fabric.nac.yaml
├─ rules/             # your own checks
├─ tests/             # test templates
├─ main.tf            # NaC module
└─ .gitlab-ci.yml     # the pipeline
```

THE PIPELINE · MR: VALIDATE, PLAN → MAIN: DEPLOY, TEST



Desired state lives in Git. The tools work out the delta.

The merge request is the change ticket: diff, plan, tests, approver.

Same stages you will see in the demo.

GitLab · GitHub Actions · Azure DevOps

nac-validate + nac-test: MPL-2.0 on GitHub

Terraform providers from the product teams

github.com/netascode · example pipelines in the public repos

cisco

PM

01 · "CAN I USE THIS IF IT'S NOT CISCO?"

Yes. The data model is yours to build.

1 **Data model**
YAML that describes intent, in a structure **you** define

2 **Schema + rules**
`nac-validate`: a Yamale schema, rules in Python

3 **Module + provider**
Terraform maps the YAML to the vendor's API

4 **Tests**
`nac-test`: Robot templates rendered from the same YAML

lb.nac.yaml · ANY VENDOR · ILLUSTRATIVE

```yaml
load_balancer:
  virtual_servers:
    - name: web-prod
      vip: 10.10.20.10
      port: 443
      pool: web-pool
  pools:
    - name: web-pool
      members: [10.10.30.11, 10.10.30.12]
```

schema.yaml · YAMALE

```yaml
load_balancer:
  virtual_servers: list(include('vs'))
---
vs:
  name: str()
  vip: ip()
  port: int(min=1, max=65535)
  pool: str()
```

Cisco's data models are **one worked example**. If a platform has an API and a Terraform provider, **the same method works**. Partners can build third-party models too.

sources: github.com/netascode/nac-validate · nac-test (MPL-2.0)

cisco

Adding is easy. Deleting is the hard part.

TEMPLATE · YOU WRITE THE STEPS

```
ntp server 10.0.0.1
ntp server 10.0.0.2

! remove .2? write another template:
no ntp server 10.0.0.2
```

You must know what is on the device **today**, and in which order to undo it.

DATA MODEL · YOU WRITE THE RESULT

```
ntp:
  servers:
    - 10.0.0.1
-   - 10.0.0.2

plan: 1 to change, removes 10.0.0.2
```

Delete the line. The tool computes the delta, **and shows it before it runs.**

A Swiss enterprise customer chose **Network as Code** over controller templates for exactly this.

PM

01 · SMALL, REVERSIBLE CHANGES

One small change at a time. Every one can be undone.

```
access_switch.nac.yaml
@@ port profile: meeting rooms @@
- vlan: 120
+ vlan: 121 # aligned to LLD
+ authentication: dot1x-mab

terraform plan: 2 to change, 0 to destroy
nac-test: 48 passed · approval: required
```

Atomic: one change per merge request

Reversible: git revert, re-apply

Reviewable: a person reads the diff

Testable: the pipeline proves it

Auditable: who, what, when, why

Repeatable: the same on all sites

cisco

PM

01 · TWO WAYS TO START

Build it yourself, or let Cisco build it with you.

NETWORK AS CODE · PUBLIC ON GITHUB

Do it yourself

- Terraform providers (product teams) + public NaC modules
- nac-validate, nac-test on GitHub
- Your pipeline, your pace; community support through GitHub issues

SERVICES AS CODE · CISCO CX

Built for you, with you

- Readiness, enablement, CI/CD setup, ITSM integration
- Brownfield import, rule + test libraries
- Enterprise support for the full setup

98%+
change success

5×
faster deployments

300+
customer deployments

13+
architectures

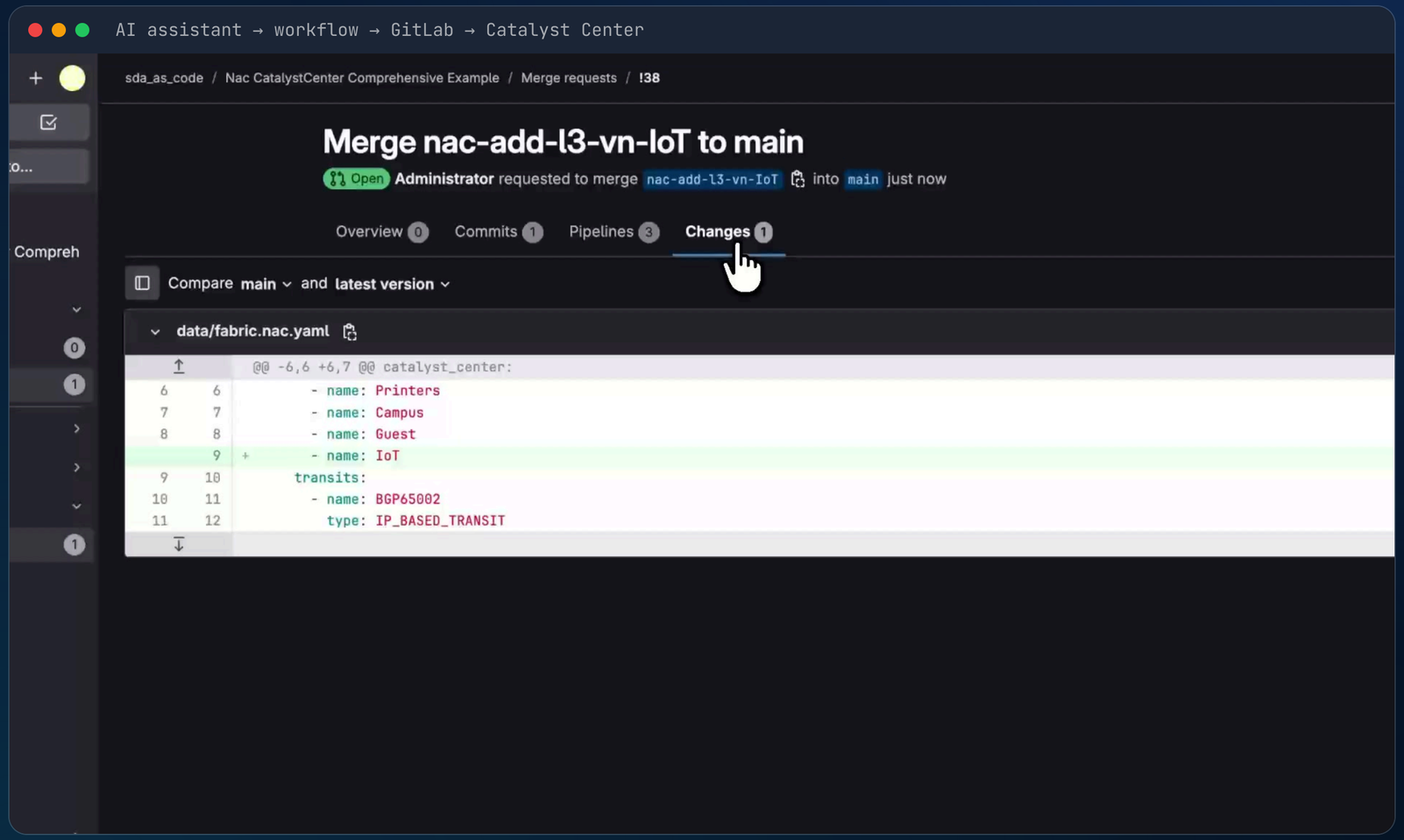
sources: Cisco NaC white paper, Mar 2026 · netascode.cisco.com

cisco

PM

02 · DEMO

From a prompt to a running network, through Git



1. **Ask**
   "Run workflow to add an L3 virtual network"

2. **Fill**
   The workflow asks: name **IoT**, action **Add**

3. **Diff**
   A merge request adds IoT to `fabric.nac.yaml`

4. **Pipeline**
   Validate, plan, merge, deploy, test

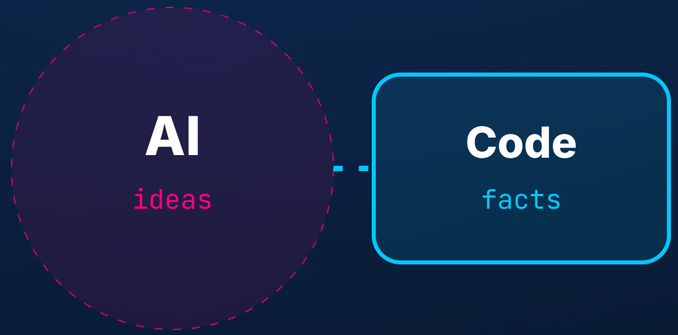
5. **Result**
   IoT is live in Catalyst Center

cisco

PM

AI is **probabilistic**.
A network must be **deterministic**.

In the demo, the AI picked a workflow, and the workflow opened a merge request: a diff to a YAML file. Nothing else. Validate, plan, review, deploy and test were the same deterministic pipeline a human change goes through, for any platform with an API.



"IaC is the foundation for AgenticOps." Cisco CX, Cisco Live 2026

cisco

DB

03 · A REAL ROLLOUT

A European insurer rebuilds its office network

~10
office sites, two countries

~350
access switches

~1,200
access points

~4,400
users

FROM

Cloud-managed switching and Wi-Fi with shared keys, third-party SD-WAN at some sites

TO

Catalyst Center + ISE (802.1X / EAP-TLS everywhere) + Cisco SD-WAN, **all as code**

Their choice: **learning by doing.** Cisco CX builds, tests and coaches; their engineers drive production.

cisco

03 · THE KEY DESIGN DECISION

Two environments, two sets of rules

LAB · THE PLAYGROUND

Auto-apply. No approval step.

- Real hardware: core pair, access stack, SD-WAN edges, WLCs, APs
- AI drafts, deploys, breaks, reads back, fixes
- Failures are cheap and teach us something

PRODUCTION · THE VAULT

Separate state, separate pipeline.

- No shared state, no symlinks: *pull, don't reference*
- Every change confirmed by a senior engineer
- Customer engineers press the button, not the AI

Let the AI fail in the lab, **so it never fails in production.**

One loop, from a spoken sentence to a tested change

A sentence in this week's call becomes an action item, a data-model change, a lab test and a result, often **before the next call.**

AI does the volume work. People hold the **decisions**, the **approvals** and the **production button**.

`Claude Code CLI agents` `several agents in parallel` `meeting bot`



What "fast" looks like when we own the loop

CORE SWITCH FAILOVER · 4 DAYS

5–6 min → **0–1**

outage, first test — lost pings, retest

Six fixes between two test days. **Every one made as code, deployed by the pipeline.**

WAN EDGE FAILOVER · SAME DAY

5:17 → **~9 s**

min outage, morning — afternoon retest

AI fused live captions + capture files into a hypothesis; the fix went through the pipeline by lunch.

Behind the scenes: capture scripts went through **60+ versions**, fix packages through **v66**. Iteration is free when the AI writes and the lab absorbs the mistakes.

Six landmines. Zero explosions.

The silent validator

Schema file had the wrong name, so it never loaded: **0 errors** reported, **129** real findings.

Unpinned provider

A provider release removed a field the module still used. Pipeline broke overnight. **Pin everything.**

The gate that never closed

The risk-score policy saw only one resource type, so approval was **always skipped**.

Lab DNA in production

Template defaults would have written **lab IP addresses** onto production devices.

Unsupported design

Planned controller cluster split across sites: not supported. **Fixing it later = reimage.**

One manual click

A rule added by hand, never in code. The as-code redeploy dropped it: **onboarding failed**.

Data access is the real accelerator

CISCO LAB · DIRECT ACCESS

Secure remote access for AI: minutes

AI can reach the lab controller and devices. It reproduced a failing template in the lab and found the cause: a description of **343 characters, limit 255**.

CUSTOMER NETWORK · HUMAN RELAY

Capture scripts: hours to days

AI writes the capture script → a person runs it from a jump host → uploads the ZIP → AI checks the hash and analyses it.

Give AI secure, read-only, audited access to the lab. It turns a day-long relay into a minute-long loop. On-prem AI needs no extra tool; Cisco has options like RADKit or Cisco IQ (see references).

read-only audited customer-controlled

DB

03 · THE HONEST PART

AI speeds up everything you control. Only that.

WHERE WE HAVE ACCESS

- Same-day fix, retest in the afternoon
- 6 fixes between two test days
- 3-hour call → actions + transcript within the hour
- Pitfalls found before production, not after

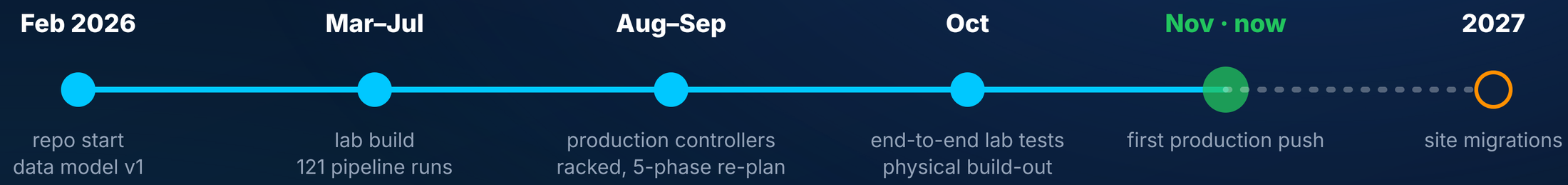
WHERE WE WAIT

- Admin rights on a test laptop: "not by test day"
- Routing owned by another team, no date
- Licences behind a proxy path not yet open
- Third-party firewall, a missing production server
- Ports reconfigured by another team, then restored

AI can't speed up a ticket in **someone else's queue.**

cisco

From empty repo to first production push



~860
commits

108
Research → Plan → Implement cycles

36
meetings transcribed

7
pipelines

04 · WHAT WE'D TELL YOU TO DO FIRST

Six lessons you can use on Monday

01
Build the lab first
A playground where AI is allowed to fail is the biggest speed-up of all.

02
No change without a diff
Code is the guardrail. One manual click will come back to bite you.

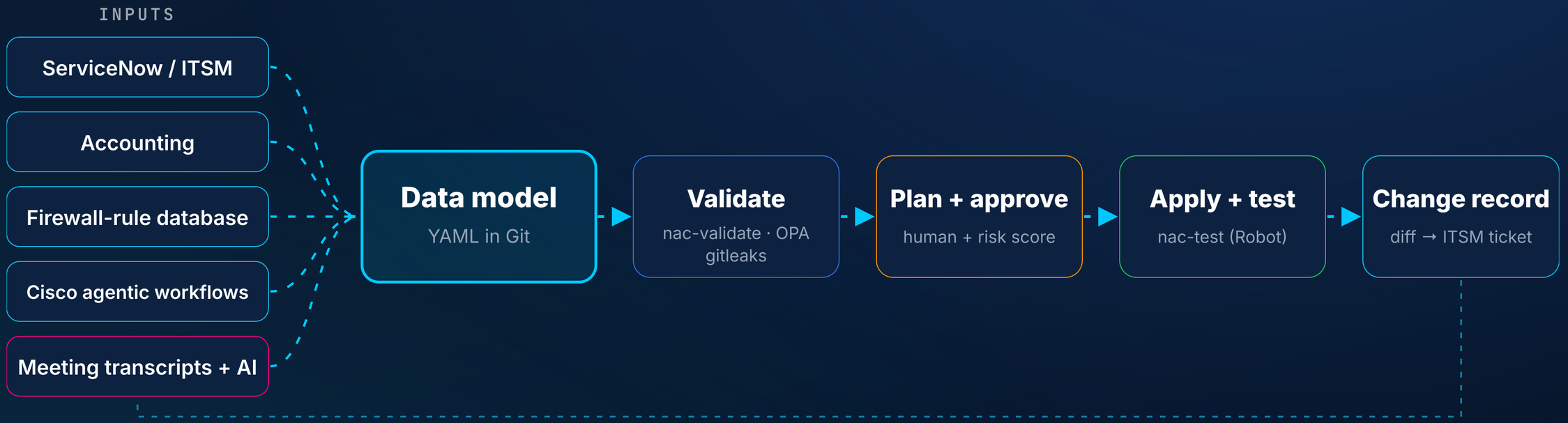
03
Test your tests
Pin versions. Check that validators and gates actually fire.

04
Give AI eyes
Safe, read-only device access closes the loop.

05
Capture every meeting
Context is AI's fuel. Transcripts in the repo, next to the code.

06
Bring the other teams
Firewall, servers, identity, licences: plan them in on day one.

The data model is the hub. Your other tools feed it.



Any source that can open a merge request on the YAML can feed the pipeline

Every apply creates a change record with the exact diff, so audit is automatic

Same gates for every input: no tool bypasses validate, review and test

QUESTIONS?

Let the AI break the lab.
Keep the network boring.

START HERE

netascode.cisco.com
github.com/netascode

TALK TO US

Patrick Mosimann · SE
Dawid Bednarczyk · CX

BY THE WAY

This deck and its rehearsal video were rebuilt by AI agents (Claude Code, Cursor) from the recordings of our planning calls.


CISCO

APPENDIX · REFERENCES

Where to go next

NETWORK AS CODE

Method, data models, docs

netascode.cisco.com

OPEN-SOURCE TOOLS

nac-validate, nac-test, modules

github.com/netascode

SECURE ACCESS FOR AI

Remote access to devices

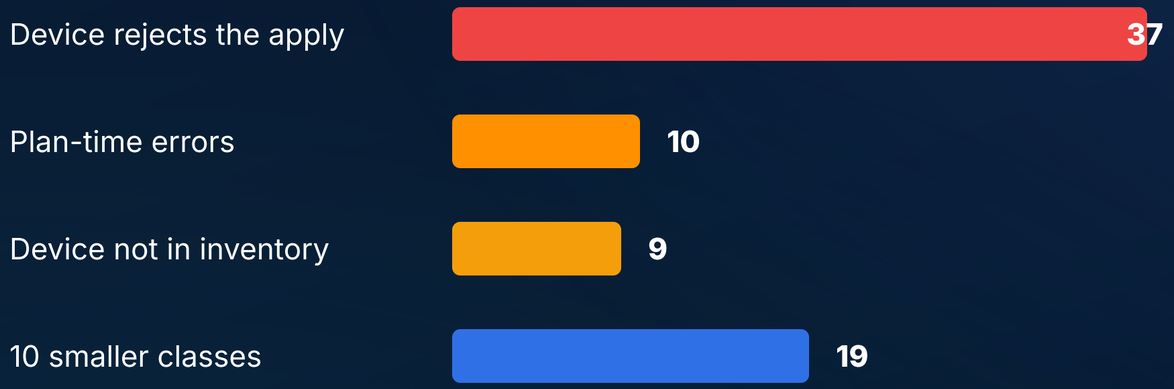
Cisco RADKit: radkit.cisco.com

Cisco IQ: iq.cisco.com

APPENDIX · LEARNING FROM 121 PIPELINE RUNS

AI read every failed run, and built the safety net

75 FAILED RUNS · BY CLASS (approx.)



46 runs passed. 13 failure classes in total.

Result: one local 8-gate pre-check that runs before anything ships.

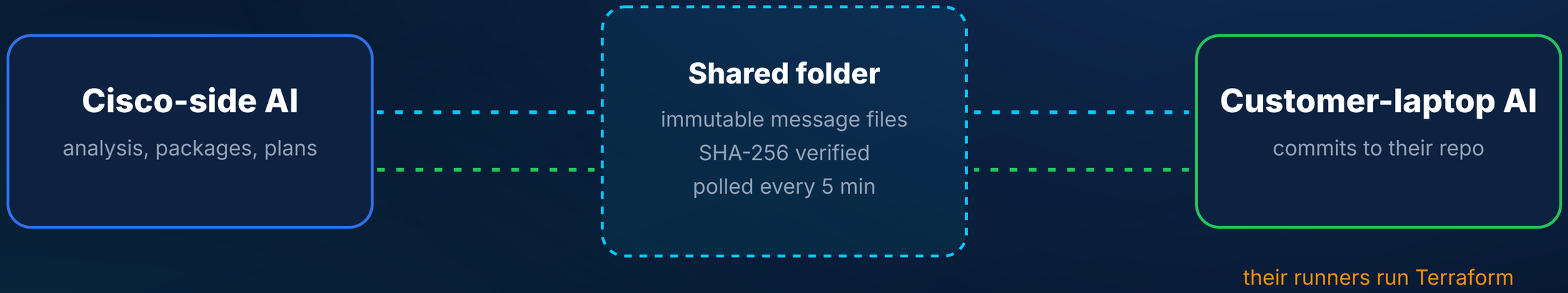
"Fixing one gate just exposes the next one. **Run the whole battery.**"

AI's own lesson, written into the repo

cisco

APPENDIX · EXTENDED · TWO AIS, ONE PROJECT

The customer's AI and our AI exchange work, safely



Before: a person carried every message by hand. Now the AIs exchange **data-model packages with their own gates**, and the customer's pipeline stays the only way into the network.

APPENDIX · HOW A CHANGE EARNS TRUST

Four tiers of testing before production

TIER 1

Review

Human + AI read the diff and the plan.

TIER 2

Schema + render

Validate the data model, render templates, no lab needed.

TIER 3

Lab deploy

Apply to the lab, read back from real devices.

TIER 4

Life of the packet

End-to-end client tests: onboarding, failover, policy.

APPENDIX · RULES THE AI WORKS UNDER

Guardrails, written down and enforced

No AI login to production

Customer pipelines apply; people approve.

AI drafts, people send

No message goes out in anyone's name without a person.

Sources for every claim

Knowledge base tracks verified / open / proposal, with source IDs.

Two channels, never mixed

Data-model packages vs capture scripts.

Destroy needs a human

No-destroy policy rule, two-person plan review, state backup.

Customer-safe outputs

Separate internal and external transcript versions.

cisco

APPENDIX · STEP 1–2 · THE BOT IN THE ROOM

One 3-hour project call, captured by a bot

1,043
caption lines, by speaker

676
screenshots of shared screens

15
topics tracked live

14
action items with owners

The advanced transcript

- Decisions, actions, open questions, risks
- A glossary of what captions mishear: `"ter form" → Terraform`
- An internal and a customer-safe version
- 36 meetings, all stored in the project repo

MEETING CHAT · CUSTOMER ENGINEER

"For **AI** action tracker update ..."

People started writing *to the AI*.

cisco

APPENDIX · AI AS PROJECT OFFICE

The plan, the tracker and the briefing are AI-built

5-phase re-plan

26 grouped items, 15 risks. The PMs adopted it into their own workbook.

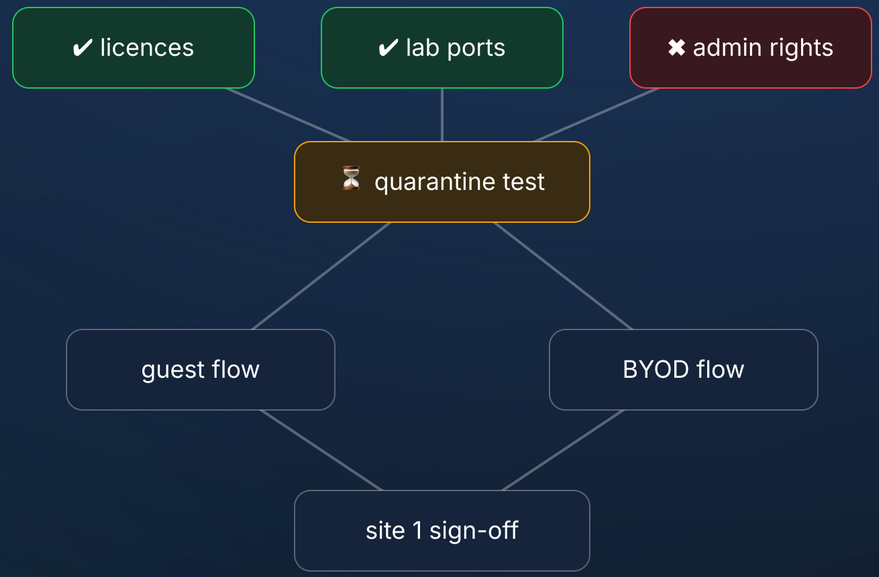
Action tracker

Each call becomes tracker updates: new, changed, closed. 141 action IDs so far.

AI drafts, people send

Every e-mail is drafted, none is sent by AI. A hard rule.

PRE-MEETING BRIEFING · DEPENDENCY MAP

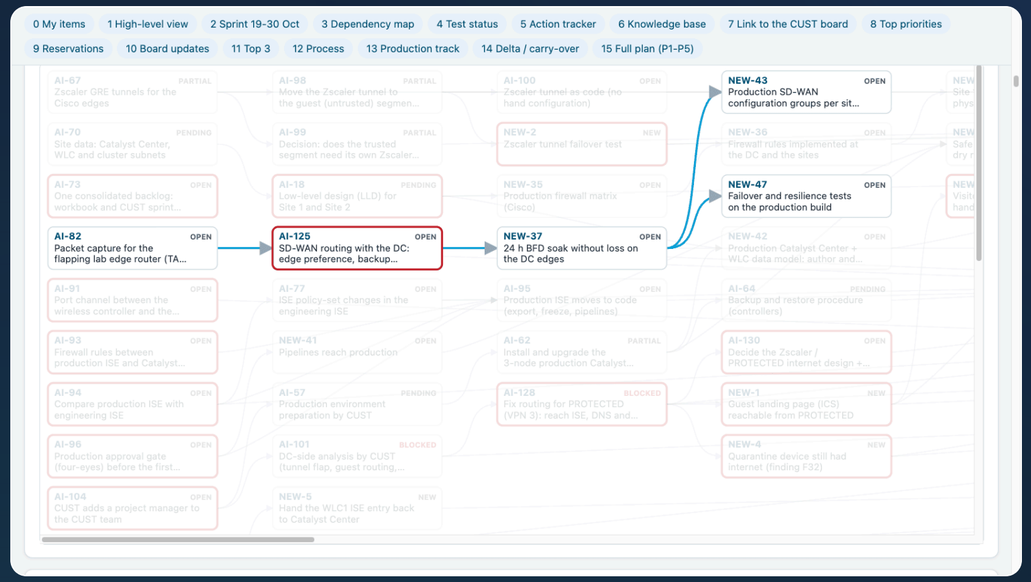


Tick what is done → see what is unblocked. (illustrative)

APPENDIX · AI DELIVERABLES · 1 OF 3

The AI keeps the plan alive: graph and tracker

DEPENDENCY GRAPH



What blocks what

Pick an item: everything before and after it lights up.

ACTION TRACKER



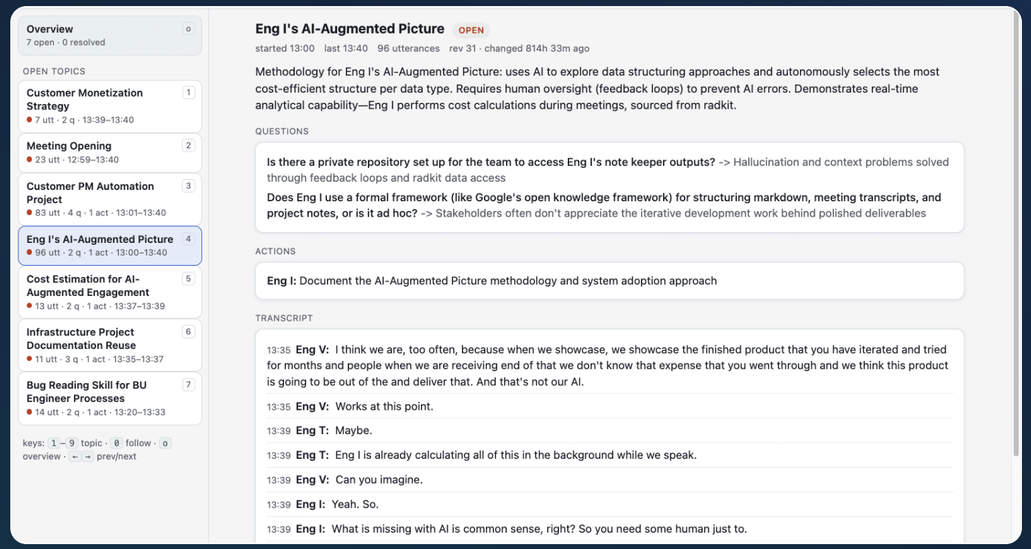
Every change, with a reason

Old and new row side by side, a side note says what, why, source.

APPENDIX · AI DELIVERABLES · 2 OF 3

The AI listens: live topics, then decisions and actions

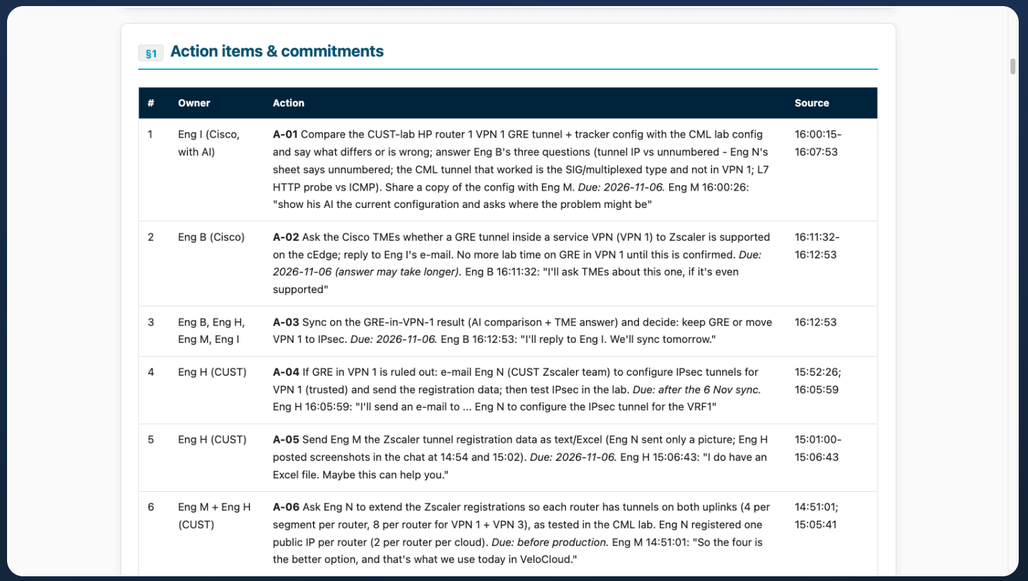
LIVE THREADS PAGE



Fills itself during the call

Topics, open questions and actions appear while people talk.

ADVANCED TRANSCRIPT



§1 Action items & commitments

#	Owner	Action	Source
1	Eng I (Cisco, with AI)	**A-01** Compare the CUST-lab HP router 1 VPN 1 GRE tunnel + tracker config with the CML lab config and say what differs or is wrong; answer Eng B's three questions (tunnel IP vs unnumbered - Eng N's sheet says unnumbered; the CML tunnel that worked is the SIG/multiplexed type and not in VPN 1; L7 HTTP probe vs ICMP). Share a copy of the config with Eng M. *Due: 2026-11-06.* Eng M 16:00:26: "show his AI the current configuration and asks where the problem might be"	16:00:15–16:07:53
2	Eng B (Cisco)	**A-02** Ask the Cisco TMEs whether a GRE tunnel inside a service VPN (VPN 1) to Zscaler is supported on the cEdge; reply to Eng I's e-mail. No more lab time on GRE in VPN 1 until this is confirmed. *Due: 2026-11-06 (answer may take longer).* Eng B 16:11:32: "I'll ask TMEs about this one, if it's even supported"	16:11:32–16:12:53
3	Eng B, Eng H, Eng M, Eng I	**A-03** Sync on the GRE-in-VPN-1 result (AI comparison + TME answer) and decide: keep GRE or move VPN 1 to IPsec. *Due: 2026-11-06.* Eng B 16:12:53: "I'll reply to Eng I. We'll sync tomorrow."	16:12:53
4	Eng H (CUST)	**A-04** If GRE in VPN 1 is ruled out: e-mail Eng N (CUST Zscaler team) to configure IPsec tunnels for VPN 1 (trusted) and send the registration data; then test IPsec in the lab. *Due: after the 6 Nov sync.* Eng H 16:05:59: "I'll send an e-mail to ... Eng N to configure the IPsec tunnel for the VRF1"	15:52:26; 16:05:59
5	Eng H (CUST)	**A-05** Send Eng M the Zscaler tunnel registration data as text/Excel (Eng N sent only a picture; Eng H posted screenshots in the chat at 14:54 and 15:02). *Due: 2026-11-06.* Eng H 15:06:43: "I do have an Excel file. Maybe this can help you."	15:01:00–15:06:43
6	Eng M + Eng H (CUST)	**A-06** Ask Eng N to extend the Zscaler registrations so each router has tunnels on both uplinks (4 per segment per router, 8 per router for VPN 1 + VPN 3), as tested in the CML lab. Eng N registered one public IP per router (2 per router per cloud). *Due: before production.* Eng M 14:51:01: "So the four is the better option, and that's what we use today in VeloCloud."	14:51:01; 15:05:41

Already sorted when the call ends

Actions with owner, due date and the minute they were said.

APPENDIX · AI DELIVERABLES · 3 OF 3

The AI briefs the team and builds the lab walkthrough

DAILY BRIEFING

CUST team

Person	Priority 1	Priority 2	Priority 3
Eng A	Point Cisco to the Site 1 Check Point firewall export AI-315 · due Wed 4 Nov	Confirm the reserved IP list for Site 1 and Site 2 (30 Oct list) AI-314 · due Wed 4 Nov	Migration plan: say yes to the Cisco-prepared draft and pick a workshop date D-175 · D-77 · due Fri 6 Nov
Eng G	Data centre firewall rule for DNS from lab VRF 1 AI-196 · D-157 · due Wed 4 Nov	Same routes on DC edge 1 and DC edge 2 (VPN 1 and VPN 99) and jump host SSH to both DC edges AI-125 · AI-149 · D-54 · due Wed 4 Nov	Choose the VRF 3 (PROTECTED) route to ISE and DNS, with Eng H and Eng B AI-128 · D-51 · D-134 · due Wed 4 Nov
Eng H	Decide the VRF 3 (PROTECTED) route to ISE and DNS (option 1 or x.x.x.x) AI-128 · D-51 · D-131 · due Wed 4 Nov	Zscaler tunnel failover: why the SIG route stays when the trackers go down AI-224 · AI-143 · due Thu 5 Nov	Production data centre hand-off and production SD-WAN edges for both sites AI-280 · AI-281 · due this sprint
Eng M	Zscaler API rights for automatic tunnels: get the answer from Eng N AI-313 · D-47 · due Wed 4 Nov	Legacy flows solved before cutover (Site 1 site-to-site flows, PRIVATE zone) AI-291 · due this week	Repeat the end-to-end test of the engineering environment with Eng C AI-319 · due this week
Eng C	ISE pipeline hand-over on the lab day (Thu 5 Nov) with Eng K D-58 · due Thu 5 Nov	Lab access switch 2: power supply B on its own feed (lab day) AI-134 · due Thu 5 Nov	Add failover, disaster-recovery and security tests to the test protocol and name the on-site person for the recordings AI-92 · AI-228 · due Fri 6 Nov
Eng L	Production DNS and NTP server addresses D-91 · due Fri 6 Nov	Production DHCP plan per VRF AI-285 · due this sprint	Production Catalyst Center backup target and procedure D-119 · AI-64 · due this sprint

Cisco team

Person	Priority 1	Priority 2	Priority 3

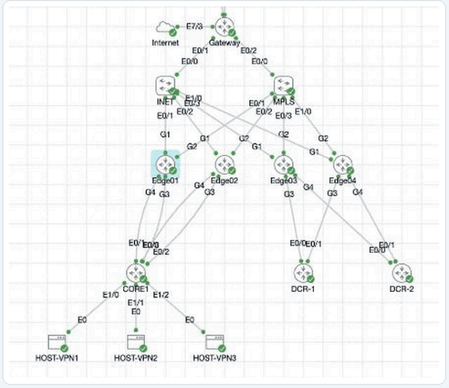
Everybody knows what to do today

Three priorities per person, each linked to its tracker item.

LAB WALKTHROUGH VIDEO

Recorded lab walkthrough (Cisco CML)

Edge router: broken like in the field, fixed in three steps, then a failover test. Frames from the 8-minute clip.



Lab topology: 4 edge routers, 2 DC routers, core, 3 VRF hosts

FIXED

All four IP SLA trackers reachable (Up)

```
Track Type        Instance              Parameter       State Last Change
516   ip sla      516                   reachability    Up    00:00:29
517   ip sla      517                   reachability    Up    01:19:09
518   ip sla      518                   reachability    Up    01:19:39
519   ip sla      519                   reachability    Up    01:19:09
```

FAILOVER

Path A blocked: trackers 516 and 518 Down, VRF 1 on tunnel 517

```
Track Type        Instance              Parameter       State Last Change
516   ip sla      516                   reachability    Down  00:00:24
517   ip sla      517                   reachability    Up    01:22:39
518   ip sla      518                   reachability    Down  00:00:29
519   ip sla      519                   reachability    Up    01:22:39
```

CLI output shown: show track brief (router prompt removed)

Source: CML walkthrough clip, 8 min 9 s, synthetic voice, AI-assisted. Customer and host names removed.

Narrated, 8 minutes, AI-assisted

Edge router broken like in the field, fixed in three steps, then a failover test.